

What Provenance Buys: A Negative Result on Label-Free Accuracy Estimation in Two-Tier Video Analytics

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Abstract

A deployed video-analytics pipeline that falls back from a learned engine to a classical one produces a mixed-provenance output stream. The operator’s question is then unavoidable: how accurate is today’s output, given no labels? We record the inference path of every record and ask whether stratifying a label-free accuracy estimator by that path beats an unstratified baseline. **It does not.** Across three engine pairs, provenance stratification and a global calibration prior are indistinguishable on whole-batch accuracy (absolute error 0.0041–0.0116 against 0.0006–0.0085 for the prior); the prior is *better* on two of three pairs. This is the expected outcome, not a bug: average confidence is provably unbiased under calibration, and our production and calibration distributions match by construction, so there is no slice shift for stratification to exploit.

What provenance does buy is the quantity an operator actually acts on, which no unstratified estimator can express at all: accuracy conditioned on inference path. We recover a primary-path accuracy of 0.6605 as 0.6613 and a secondary-path accuracy of 0.4267 as 0.4699 on the object pair, and the analogous split on the other two. We also report a systems result that survives independently of the estimation question: the two-tier pipeline holds yield at 1.0 up to an injected primary-failure rate of 0.5, where a primary-only pipeline falls to 0.553, and the classical secondary is bit-for-bit reproducible across processes (0 mismatches in 5,000 records per pair). We argue the honest claim is the per-path estimate and the bounded-latency determinism, not a new estimator, and we name the baseline that beat us.

All numbers in this paper are produced by behavioural engine models on synthetic traces with fixed seeds. They are labelled SYNTHETIC throughout and none of them is a measurement of a production YOLO, FaceNet or PaddleOCR stack.

1 Introduction

Video-analytics platforms in the field do not run one model per task. They run a learned primary engine and, when it times out, crashes, or fails a health probe, a classical secondary: a Haar cascade behind a CNN face matcher, a template matcher behind a learned plate reader, a background-subtraction blob detector behind an object detector. The pipeline is designed so that a frame is never silently dropped. The consequence is a mixed output stream in which some records came from a strong model and some from a weak one, and the mix changes minute to minute with the health of the primary.

An operator staffing a control room asks a simple question about that stream: *how much of what I am looking at right now is right?* There are no labels. The obvious answer is to average the model’s confidence over the batch. The tempting answer, once the pipeline already records

which engine produced each record, is to stratify: estimate accuracy per inference path from a labelled calibration set, then reweight by today’s observed path mix.

This paper reports that the tempting answer does not beat the obvious one, states why it cannot under the conditions we and most deployments operate in, and identifies the narrower claim that does survive.

1.1 The negative result, stated plainly

Let $\hat{a}_{\text{prior}}$ be the global calibration prior: the accuracy measured on a held-out labelled set, applied unchanged to production. Let $\hat{a}_{\text{prov}}$ be the provenance-stratified estimator: per-path accuracies from the same calibration set, reweighted by the production path mix. On three engine pairs with 3,000 calibration and 17,000 production records each, the absolute errors are 0.0041, 0.0116 and 0.0054 for $\hat{a}_{\text{prov}}$ against 0.0006, 0.0085 and 0.0061 for $\hat{a}_{\text{prior}}$. The prior wins on two of three, and the differences are inside the 95% interval on the true accuracy itself. Confidence-only estimation is worse than both (0.0105–0.0140) because our engines are imperfectly calibrated (ECE 0.0127–0.0167), which is a property of the engine models, not a defence of stratification.

Kivimäki et al. [11] give the reason. Average confidence is an unbiased estimator of accuracy whenever the scores are calibrated, with no distributional assumption beyond that. Stratification helps only when the calibration and production distributions differ *within* the strata used — the slice-shift setting Chen et al. [3] were built for. Our production stream is drawn from the same generator as our calibration set. There is no shift. There is therefore nothing for provenance to correct, and it would be surprising if it won.

1.2 Contributions

1. A negative result: provenance stratification does not improve whole-batch label-free accuracy estimation in a matched-distribution two-tier pipeline, with the theoretical reason for why not (§3).
2. The narrower positive claim that does hold: per-path accuracy is recoverable from the same machinery and is not expressible by any unstratified estimator (§4), together with an honest report of where that recovery is loose (secondary-path error up to 0.043).
3. A systems result independent of the estimation question: the two-tier pipeline trades accuracy for bounded latency and cross-process determinism, and loses to a retry baseline on raw accuracy (§5). We report the loss rather than omit the baseline.

2 Setting

2.1 The deployed two-tier pipeline

Each analytic task is a pair (primary, secondary). We study three: object detection, face matching, and automatic number-plate recognition. The dispatcher calls the primary under a deadline. On timeout, exception, or a failed health probe it calls the secondary and marks the record. The secondary is always called; there is no drop path.

2.2 The record

Every emitted record carries nine fields, of which four matter here: the entity key, the engine confidence, the `source` field naming the engine that produced it, and the `path` field taking the value `primary` or `secondary`. Provenance is therefore free at estimation time — it is already written for audit reasons — which is exactly why the stratified estimator is worth testing and exactly why a negative result is worth publishing rather than quietly dropping.

Pair	True	Prov. strat.	Global prior	Conf. only
objects	0.6352	0.6406 (0.0054)	0.6413 (0.0061)	0.6247 (0.0105)
faces	0.5949	0.5833 (0.0116)	0.5863 (0.0085)	0.5808 (0.0140)
anpr	0.5794	0.5835 (0.0041)	0.5800 (0.0006)	0.5676 (0.0118)

Table 1: **Whole-batch accuracy estimation (synthetic).** Absolute error in parentheses. $n_{\text{cal}} = 3,000$, $n_{\text{prod}} = 17,000$. The global prior is better than provenance stratification on two of three pairs and never worse by more than 0.003. **This table is the negative result.**

Pair	Primary path		Secondary path		ECE
	True	Estimate	True	Estimate	
objects	0.6605	0.6613	0.4267	0.4699	0.0154
faces	0.6328	0.6207	0.3647	0.3557	0.0127
anpr	0.6155	0.6248	0.4053	0.3844	0.0167

Table 2: **Per-path accuracy (synthetic).** The gap between paths is 0.21–0.27 and is recovered to within 0.001–0.043. No unstratified estimator produces these columns at all. The secondary-path estimate on the object pair is the loosest (0.043); we do not claim uniform tightness.

2.3 Threat to validity we accept up front

The engines here are behavioural models parameterised by a difficulty scale, not the real stacks. The determinism result transfers to any pipeline whose secondary is a fixed-function classical algorithm. The accuracy numbers do not transfer and should not be read as characterising YOLO, FaceNet or PaddleOCR.

3 Why the obvious estimator wins

Write $Y \in \{0, 1\}$ for record correctness, S for the model score, and $P \in \{\text{primary}, \text{secondary}\}$ for the path. Calibration means $\mathbb{E}[Y \mid S = s] = s$. Then

$$\mathbb{E}[\bar{S}] = \mathbb{E}[\mathbb{E}[Y \mid S]] = \Pr(Y = 1),$$

so average confidence is unbiased for batch accuracy without reference to P at all [11]. Conditioning on P can reduce variance, but the variance of a mean over 17,000 records is already far below the bias we are trying to detect: the 95% interval on true accuracy is $[0.6287, 0.6432]$ for the object pair, a width of 0.0145, which is wider than every estimator gap we measure.

Stratification pays only under shift. Let w_p be the calibration path share and w'_p the production share. The prior errs by $\sum_p (w'_p - w_p) a_p$ where a_p is per-path accuracy; the stratified estimator cancels exactly this term. Our generator holds $w'_p \approx w_p$, so the term it cancels is near zero and the estimator inherits only the extra estimation noise of two smaller calibration strata. That is the mechanism behind Table 1, and it is a property of the deployment, not of the idea. A deployment whose primary degrades diurnally — night-time face failures, monsoon plate failures — has real $w'_p \neq w_p$ and should expect the opposite result. We could not manufacture that shift honestly without also manufacturing the accuracy conclusion, so we report the matched case and say so.

4 What stratification does buy

Table 2 is the case for keeping provenance. The operational question is rarely “what is batch accuracy” and usually “is the degraded path carrying enough of tonight’s output that I should

Fail rate	Two-tier yield	Two-tier acc.	Primary-only yield	Primary-only acc.	Retry acc.
0.05	1.000	0.6353	0.9309	0.6073	0.6496
0.10	1.000	0.6354	0.8919	0.5892	0.6508
0.20	1.000	0.6274	0.8023	0.5349	0.6450
0.35	1.000	0.5897	0.6782	0.4491	0.6214
0.50	1.000	0.5698	0.5530	0.3715	0.5934

Table 3: **Yield and accuracy against injected primary-failure rate, object pair (synthetic, $n = 10,000$ per row).** Two-tier never drops a frame. It also never beats retry on accuracy-over-all. Both facts belong in the paper.

Pair	Mean (ms)	p99 (ms)	Secondary share	Accuracy
objects	41.7	47.2	0.108	0.6281
faces	55.4	58.1	0.144	0.5921
anpr	81.8	100.0	0.171	0.5830

Table 4: **Latency of the fallback path (synthetic).** The p99 is bounded by construction: a fallback is one extra call with a known cost, whereas a retry schedule has an unbounded tail.

stop acting on it”. The paths differ by 0.21 to 0.27 in accuracy. An unstratified estimator reports a single number that is a weighted average of two very different populations and is therefore correct about a quantity nobody needs.

We report the loose case rather than the mean of the four: the object-pair secondary estimate is off by 0.043, roughly 10% relative. The secondary stratum is the smaller one, and its calibration sample is correspondingly thin. A deployment that wants tight per-path numbers must size the calibration set per stratum, not per batch. That is a design consequence, and it is the useful sentence in this section.

5 The systems result, and the baseline that beats it

Three things are true at once in Table 3, and papers in this area usually report only the first.

Yield. The two-tier pipeline emits a record for every frame at every injected failure rate up to 0.5. A primary-only pipeline degrades linearly to 0.553. If the downstream consumer is a tracker that needs an observation per frame, this difference is the whole system.

Determinism. The classical secondary produced identical output across independent processes on all three pairs: 0 mismatches in 5,000 records each. This is not automatic. Our first implementation keyed off Python’s builtin `hash()`, which is salted per process, so two runs of the same script disagreed while the unit tests passed. Stable hashing (`zlib.crc32`) fixed it. Anything that must reproduce across processes cannot hash strings with `hash()`; we record the trap because it is the kind of defect that survives a green test suite.

The loss. Retrying the primary beats the two-tier pipeline on accuracy-over-all at every failure rate we tested, by 0.014 to 0.032. This is not surprising — a successful retry returns a primary-quality record where the fallback returns a secondary-quality one — but it is decisive for how the contribution should be phrased. The fallback is defensible where a repeated call is *not* free and the failure is *not* transient: where the primary is down rather than flaky, where the deadline is hard, or where the retry would cost a GPU slot that another camera needs. Its wins are bounded latency (Table 4) and reproducibility, not accuracy. A paper that omits the retry column is making a claim its own data does not support.

6 Baselines

We compare against average confidence [11], the global calibration prior, difference-of-confidences [8], and the average threshold confidence family [6], both global and per-stratum. Slice-based estimation [3] is the closest prior work and is the correct comparison for the stratified estimator: it formalises exactly the reweighting we perform, with provenance as the slicing function. Our result is that under matched distributions this reduces to the prior, which is consistent with what Chen et al. [3] claim rather than in tension with it.

Per-stratum ATC ties with provenance stratification on the per-path metric to within 0.01 on all three pairs, which we state here rather than in an appendix. The honest reading is that *recording the path* is what buys the per-path number; the particular estimator layered on top of it is not the contribution.

7 Related work

The two-tier architecture is anticipated. NoScope [10] cascades a cheap specialised model in front of an expensive one with a difference detector and a learned threshold. Neural Simplex [14] switches from a learned controller to a verified fallback under a safety monitor. BranchyNet [16] exits early on confident samples. IDK cascades [17] formalise the abstain-and-defer step, and the selective-prediction line [4, 7] gives its decision theory. FrugalGPT [2] is the same pattern under a budget. We add nothing architectural to this literature and say so.

Label-free accuracy estimation is likewise a developed field: ATC [6], agreement-based estimation [1], AutoEval [5], DoC [8], projection norm [18], confidence optimal transport [12], and the older estimation-without-labels line [15]. Calibration quality is the binding constraint on all of them [9, 13], which is why our confidence-only column is the weakest.

Where we sit is therefore narrow: we test a specific combination (provenance-as-slice on a two-tier video pipeline) that the literature makes plausible, find it does not pay under matched distributions, and report the conditions under which it would.

8 Limitations

1. The engines are behavioural models with a difficulty parameter. Accuracy figures characterise the generator.
2. Failure is injected at a fixed rate, independently per frame. Real primary failures are bursty and correlated with the conditions that also make frames hard, which would create the very shift that would make stratification win.
3. The determinism claim is about the secondary engine’s output given a frame. It is not a claim that the two-tier system as a whole is deterministic; which path a frame takes depends on wall-clock timeouts.
4. Single estate, single generator, three pairs. Nothing here is validated against operational footage.

9 Conclusion

Recording the inference path of every record is cheap, is already required for audit, and does not improve label-free estimation of whole-batch accuracy. Under calibration and matched distributions the unstratified prior is unbiased and the stratified estimator has nothing left to correct; our measurements agree with the theory to within the confidence interval on the target quantity. What the path field does buy is the per-path accuracy split, which no unstratified

estimator can produce and which is the number an operator acts on. Separately, the two-tier pipeline buys yield and cross-process determinism at a measured cost in accuracy-over-all relative to retry. We recommend that deployments record provenance, report accuracy per path, and stop expecting the batch number to improve.

Reproduction

```
cd code
pip install -r requirements.txt
python -m pytest -q
python -m prresearch.p2_fallback.experiment
```

Fixed seeds; `results/p2_fallback.json` is byte-identical across runs and across processes. Code and data: <https://github.com/amitduabits/PRAHARI-P2-NegativeResult>

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